

GV Agent Safety V2

Experimental Runtime Governance for Agentic Systems
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Research question: Can protected constraints and recoverability boundaries remain enforced when an optimizing agent is rewarded for crossing them?

What it is

GV Agent Safety V2 is an experimental runtime-governance architecture. Untrusted work executes in disposable state behind selected operating-system restrictions. The system evaluates observed effects together with protected invariants and Sentinel recoverability state before deciding whether results may enter authoritative state. Reward or task success does not automatically override a protected boundary.

Architecture

Agent objective	→	Proposed action	→	Disposable execution
				↓
Authoritative state	← COMMIT	ALLOW / DENY / HALT	←	Observed effects + Sentinel

Current prototype boundary

Linux enforcement • Bubblewrap namespace isolation • read-only protected safety mounts • isolated network namespace • dropped Linux capabilities • libseccomp denial of tested child-process creation syscalls	Governance • protected action/effect invariants • effect-based decisions rather than action names alone • disposable transactional execution • Sentinel stable / warning / critical / irrecoverable states • authoritative commit only after GV approval
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Reproducible evidence

Evidence	Result
Full automated suite	95 tests passed
GitHub Actions	Clean CI run on pull request
Consolidated V2 acceptance suite	4 tests passed
Machine-readable acceptance report	passed: true; exit code 0
Permitted transaction	safe_work allowed and result.txt committed
Protected-file attempt	denied; not committed
External-network attempt	denied; not committed
Child-process creation attempt	denied; not committed
Combined adversarial transaction	all selected prohibited effects denied; no commit

Scope and non-claims

Tested claim: In the tested Linux/Bubblewrap environment, selected protected filesystem, network, process-creation, and authoritative-commit boundaries remained enforced during the included adversarial action tests.	Not claimed: general AI alignment; universal containment; resistance to arbitrary hostile code; protection against kernel/Bubblewrap/libseccomp vulnerabilities; or proof that an AI system is safe.
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What I am looking for

Adversarial technical review. I am specifically interested in attempts to falsify the boundary: malformed worker output, sandbox crashes, commit-path/symlink attacks, resource exhaustion, inherited descriptors, recoverability-state edge cases, and other ways the prototype could fail closed incorrectly or allow an unsafe result to become authoritative.

Repository: github.com/willshacklett/gvai-safety-systems
Prototype tag: `gv-agent-safety-v2-prototype` **Implementation commit:** `f5aff4e`
Research rule: "Add a claim only after an experiment can falsify it."